Muhammad Shahbaz

Full-Stack Engineer · AI/ML

Karachi, Pakistan · shahbaz21042005@gmail.com · +92 336 3600215 · shahbaz2104.github.io · github.com/Shahbaz2104

Full-stack engineer with 1+ year of professional experience building production web applications and teaching modern development. I ship end-to-end products with Next.js, React, TypeScript, Node.js, PostgreSQL, and Prisma, and I build ML pipelines with Python and scikit-learn. Comfortable owning schema design, APIs, authentication, payments, testing, and deployment. Looking for full-stack or AI/ML engineering roles where product quality and honest metrics matter.

EXPERIENCE

Programming Instructor | Aptech

Jun 2025 – Oct 2026

- Delivered project-based training in JavaScript, TypeScript, React, Node.js, Express, PHP, and Laravel to 100+ beginner and intermediate students.
- Taught database design, SQL, and the full SDLC — planning, UI/UX, system design, testing, and deployment — using production workflows.
- Reviewed student code for architecture, debugging, testing, and Git hygiene; designed capstone projects that simulate client briefs.

SELECTED PROJECTS

Titan Fitness — AI-Powered Gym Management Platform

Live · Source

Next.js · React · TypeScript · Prisma · PostgreSQL · Stripe · Better Auth · OpenAI

- Built 16 member dashboard pages, 11 admin pages, and 95+ REST API routes with role-based access control and OTP/Google authentication.
- Integrated Stripe checkout, coupons, and webhooks; added AI coaching, workout generation, and nutrition planning with rule-based fallbacks.
- Shipped a responsive PWA with SEO metadata, push notifications, and admin tools for plans, members, classes, and content.

MechSense — Industrial Machine-Health Monitoring

Source

FastAPI · PostgreSQL · SQLAlchemy · React 19 · TypeScript

- Built a FastAPI/PostgreSQL backend for threshold detection, anomaly lifecycle, work orders, and CSV ingestion, covered by 124 backend tests.
- Built a React operations console for machine health, sensors, alarms, and work orders, with a typed API client and responsive layouts.

ASHRAE Energy Prediction — ML Pipeline

Dashboard · Source

Python · pandas · scikit-learn · Streamlit

- Modeled hourly energy demand on a 1.03M-row, 120-building subset with a chronological train/validation/test split and leakage controls.
- Compared Linear Regression, Random Forest, and HistGradientBoosting; selected HistGradientBoosting at test R^2 0.970 and MAE 33.20.
- Reported the critical limitation: R^2 −0.555 on unseen buildings, with the failed generalization documented instead of hidden.

SELECTED PROJECTS, CONTINUED

Hospital Management System — Enterprise Operations Platform

Live · Source

Next.js · TypeScript · Prisma · MongoDB · JWT · Stripe · Cloudinary

- Built a 27-module platform covering patients, appointments, pharmacy inventory, billing, insurance claims, HR/payroll, and role-based dashboards.
- Implemented transactional billing and stock workflows, PDF/Excel/QR exports, Stripe payments/refunds, audit logs, and RBAC for 9 roles.
- Added Cloudinary uploads for medical records and hospital branding with purpose-based size and file-type limits.

BIOSAF Enterprises — Client Website & Admin Dashboard

Live · Source

Next.js · TypeScript · Prisma · MySQL · JWT · Zod

- Delivered a production marketing site, searchable product catalog, and JWT-protected admin dashboard with CRUD for products, categories, blogs, projects, testimonials, FAQs, and inquiries.
- Improved SEO, accessibility, and load performance; diagnosed and fixed an App Router rendering bug verified with Playwright.

ADDITIONAL PROJECTS

PulseWatch — Self-hosted uptime monitor; Rust/Axum + SQLite API, React console, SSRF/DNS hardening, 187 tests.

Nihongo Lab — Japanese learning platform; Next.js/Prisma, 8,334 vocabulary, 2,211 kanji, SM-2 engine, 73 tests.

Expense Tracker Pro — Finance tracker; Rust/Axum + React, CSV import/export, exact decimal money, 164 tests.

SehatSathi AI — Cross-platform health assistant; React Native/Expo, Firebase, OpenAI chat and report summaries.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Rust, PHP, HTML/CSS

Frontend: React, Next.js, React Native, Tailwind CSS, TanStack Query, Redux, Zustand, Framer Motion

Backend: Node.js, Express, FastAPI, Laravel, REST APIs, JWT, Better Auth, Prisma, SQLAlchemy

Data & ML: PostgreSQL, MongoDB, MySQL, SQLite, pandas, scikit-learn, Streamlit

Integrations: OpenAI API, Stripe, Cloudinary, Firebase, WebSockets, Resend

Testing & Tools: Vitest, Jest, Supertest, Playwright, Zod, Git/GitHub, Docker, Vercel, Postman

Languages spoken: English (fluent), Urdu (native), German (A2), Japanese (N5)

EDUCATION

B.E. Mechanical Engineering · NED University of Engineering and Technology — Oct 2023 – Jun 2027 (in progress)

Advanced Diploma in Software Engineering · Aptech — Nov 2022 – Jan 2025